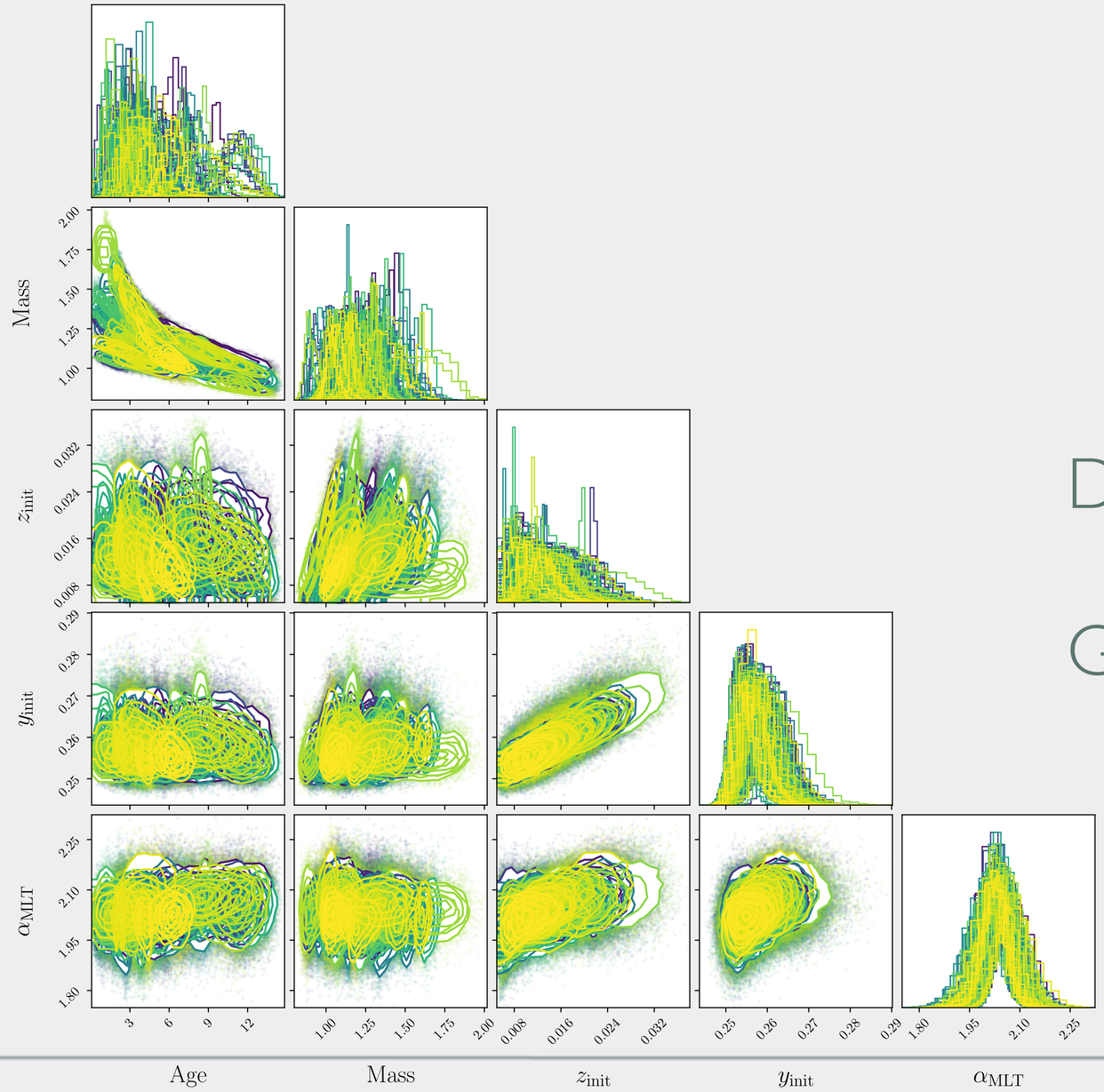
Inference from Scientific Data



Distributions Lecture

Guy Davies

Inference

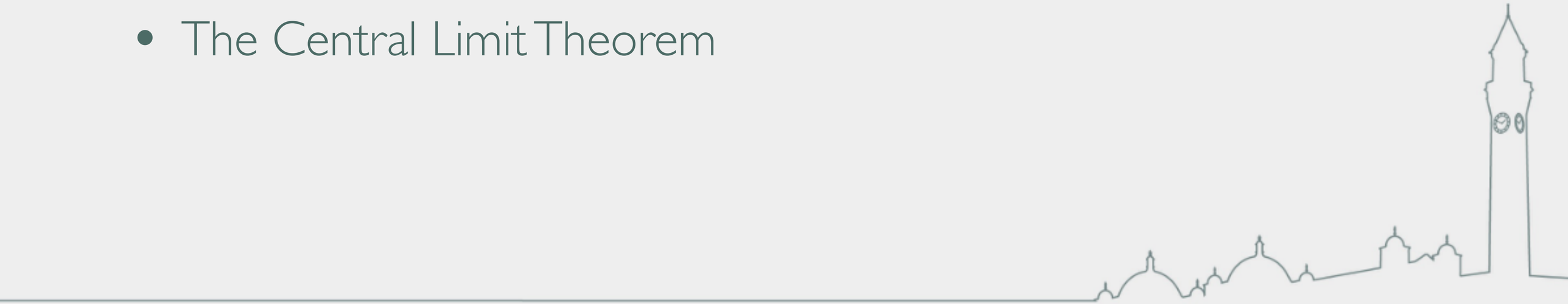
- **Introduction** - Intro, Logic, Probability recap
- **Probability** - Distributions, Moments, Central Limit Theorem
- **Bayesian methods** - Parameter estimation, Model selection, Stochastic sampling
- **Frequentist methods** - Model fitting, Hypothesis testing
- **Fun stuff** - We'll see, but maybe Gaussian Processes, …



Inference -Week 3

Today:

- Common Distributions
- Multi-variate distributions
- Marginal/conditional distributions
- The Central Limit Theorem



Inference - Week 3

The Binomial distribution

Number of successes in n experiments with a pass or fail outcome.

PMF:

$$P(r; n, p) = p^r (1-p)^{n-r} \binom{n}{r},$$

where $P(\text{pass}) = p$ and

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$



Inference - Week 3

The Poisson distribution

Random variable n is the number of events that occur when events occur independently and at a constant average rate λ. Note, there is no number of experiments here.

PMF:

$$P(n;\lambda) = \frac{e^{-\lambda}\lambda^n}{n!},$$

but be careful, as there is a difference between the rate λ and the events per unit time $\tilde{\lambda}$.



Inference - Week 3

Poisson distribution example

It is common in astronomy to need to know how long to integrate for when observering a distant object. Using what we know of a source, we can estimate how many photons per second should be arriving at the aperture of our telescope. What if we want to know how long to observe to have at least a 50% chance we get at least M photons?

PMF:

$$P(n;\lambda) = \frac{e^{-\lambda}\lambda^n}{n!},$$

$$P(n \geq M;\lambda) = \sum_{n=M}^{\infty} \frac{e^{-\lambda}\lambda^n}{n!},$$

$$P(n \geq M;\lambda) = \sum_{n=M}^{\infty} \frac{e^{-(\tilde{\lambda}T)}(\tilde{\lambda}T)^n}{n!},$$

where T is our integration time, and $\tilde{\lambda}$ is the number of photons arriving per second.

Now we could have a go at solving this analytically but I am a fan of numerical solutions. So we need a few functions. You also need a little bit of luck. Rather than relying on luck, we have some tricks!



Inference - Week 3

Trick

Probability to the rescue!

$$P(n \geq M; \lambda) + P(n < M; \lambda) = 1,$$

$$P(n < M; \lambda) = \sum_{n=0}^{M-1} \frac{e^{-(\tilde{\lambda}T)}(\tilde{\lambda}T)^n}{n!},$$

$$P(n \geq M; \lambda) = 1 - P(n < M; \lambda),$$

$$P(n \geq M; \lambda) = 1 - \sum_{n=0}^{M-1} \frac{e^{-(\tilde{\lambda}T)}(\tilde{\lambda}T)^n}{n!}.$$



Inference - Week 3

In [49]:

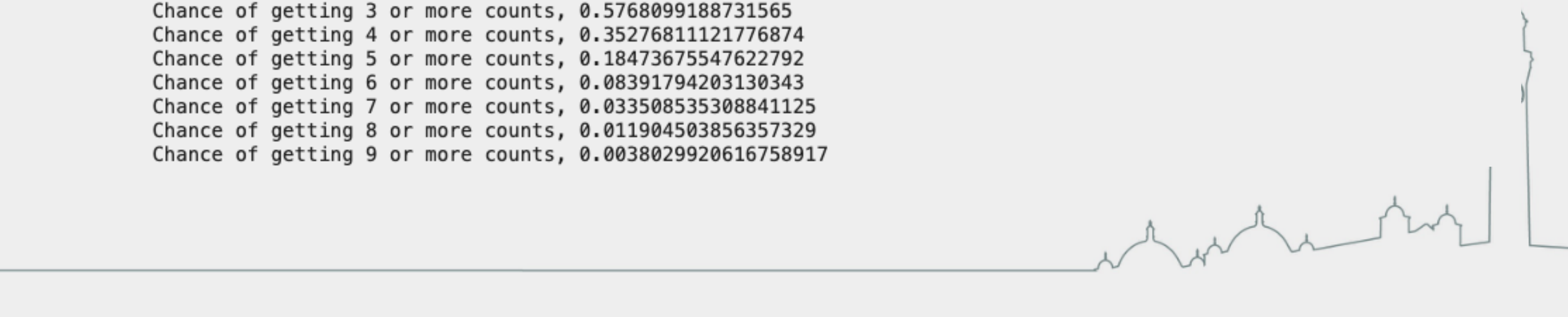
```python
from math import factorial
import numpy as np

def PMF(tlambda: float, T: float, n: int) -> float:
    return np.exp(-tlambda * T) * (tlambda * T)**n / factorial(n)

def sum_PMF(M: int, tlambda: float, T: float) -> float:
    return 1 - np.sum([PMF(tlambda, T, n) for n in range(M)])

tlambda = 3. # average counts per second expected
T = 1. # integration time
for i in range(10):
    print(f' Chance of getting {i} or more counts, {sum_PMF(i, tlambda, T)}')
```

```
Chance of getting 0 or more counts, 1.0
Chance of getting 1 or more counts, 0.950212931632136
Chance of getting 2 or more counts, 0.8008517265285442
Chance of getting 3 or more counts, 0.5768099188731565
Chance of getting 4 or more counts, 0.35276811121776874
Chance of getting 5 or more counts, 0.18473675547622792
Chance of getting 6 or more counts, 0.08391794203130343
Chance of getting 7 or more counts, 0.033508535308841125
Chance of getting 8 or more counts, 0.011904503856357329
Chance of getting 9 or more counts, 0.0038029920616758917
```



Inference - Week 3

The Gaussian distribution

Note this is a continuous distribution. I might also call this the normal distribution.

PDF:

$$P(x;\mu,\sigma) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\frac{-(x-\mu)^2}{2\sigma^2},$$

where $-\infty < x < \infty$.

There is a standard integral that will sometimes help you:

$$\int_{-\infty}^{\infty} \mathrm{d}x \exp\frac{-x^2}{2\sigma^2} = \sqrt{2\pi\sigma^2}.$$

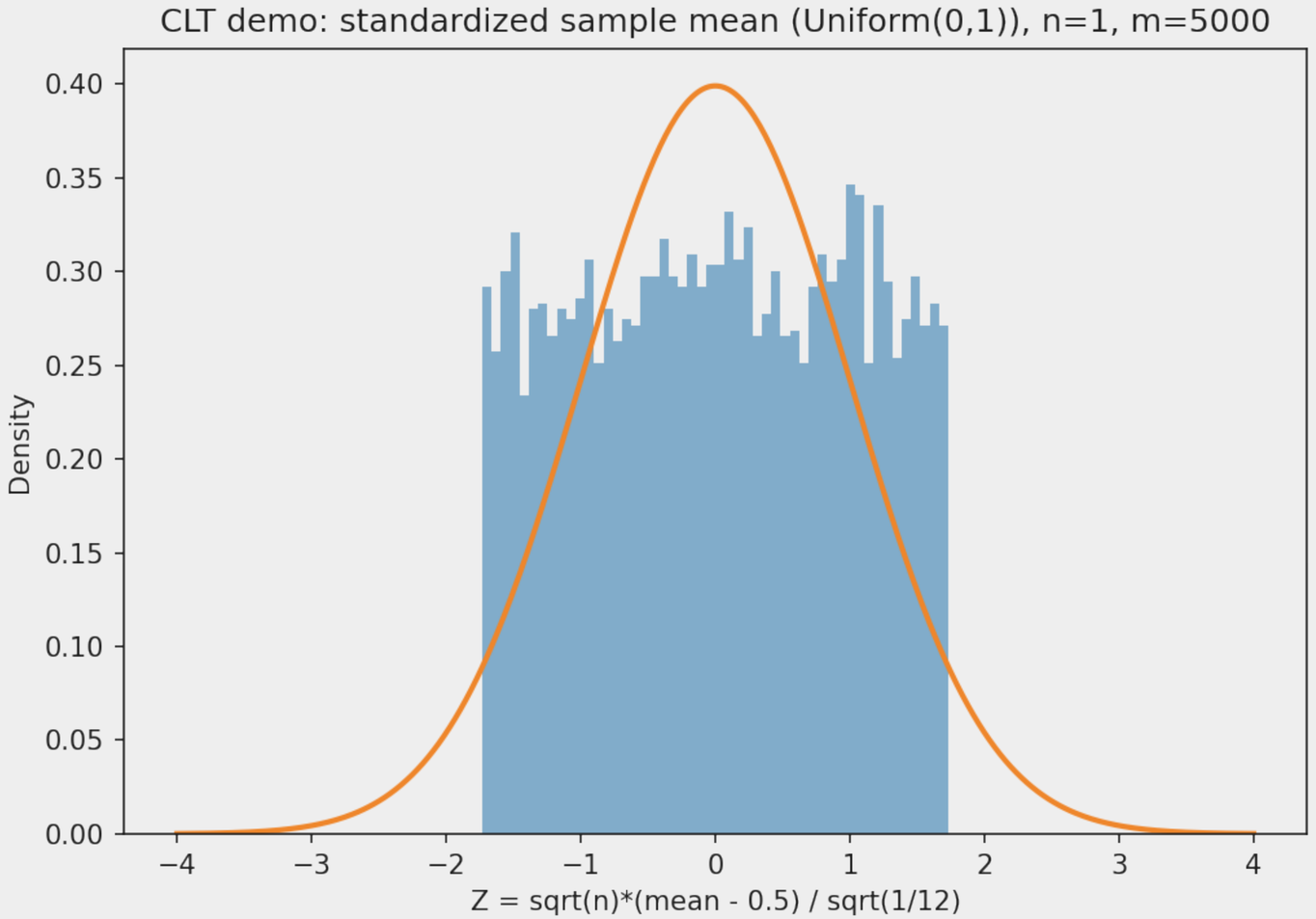
Note, a more Bayesian notation and convientient way of writing this would be:

$$x \sim \mathcal{N}(\mu, \sigma^2),$$

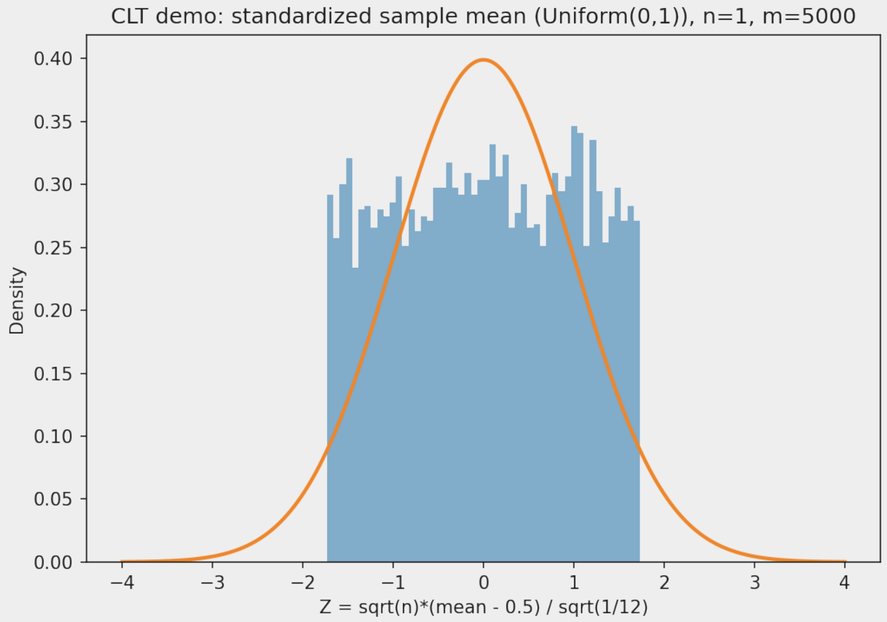
which really means that we consider x to be a random variable drawn from (or distributed according to) the normal distribution with the given mean and variance.



Inference - Week 3 CLT



Inference - Week 3 CLT




```python
rows = []
xgrid = np.linspace(-4, 4, 400)

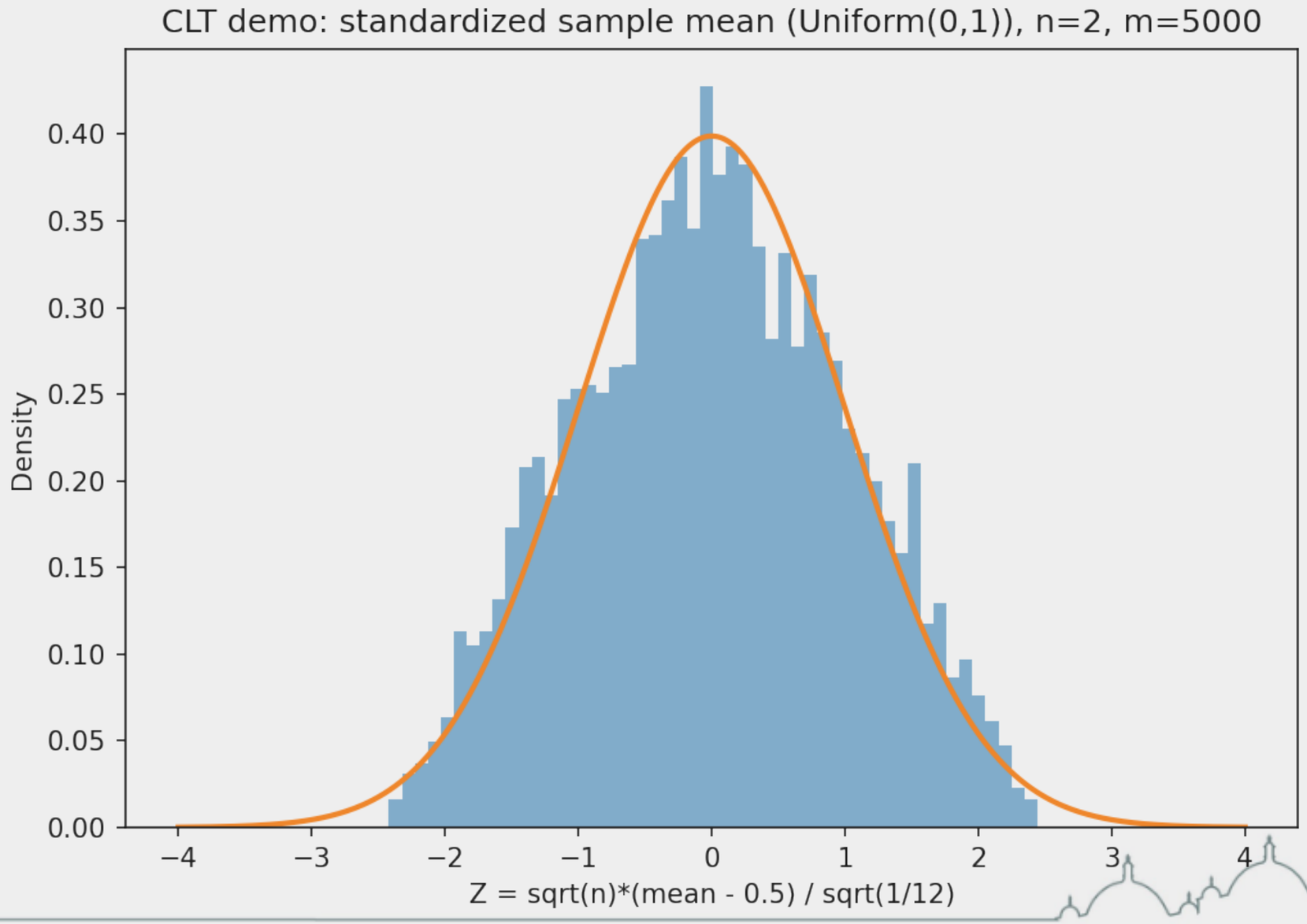
for n in n_list:
    # Simulate: shape (m, n), mean along axis=1
    samples = rng.random((m, n))
    sample_means = samples.mean(axis=1)
    # Standardize: Z ~ approx N(0,1) by CLT
    Z = np.sqrt(n) * (sample_means - mu) / sigma

    # Empirical stats
    z_mean = Z.mean()
    z_var = Z.var(ddof=1)

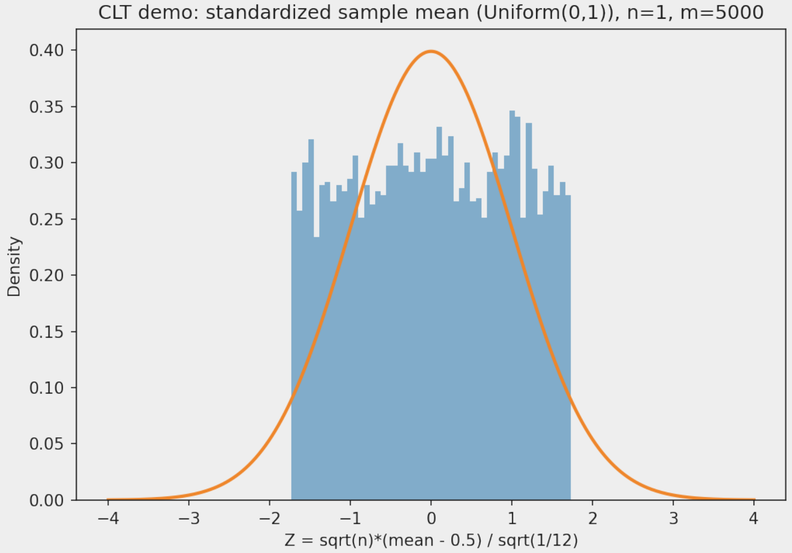
    rows.append({
        "n": n,
        "empirical_mean_Z": z_mean,
        "empirical_var_Z": z_var,
        "theoretical_mean_Z": 0.0,
        "theoretical_var_Z": 1.0
    })

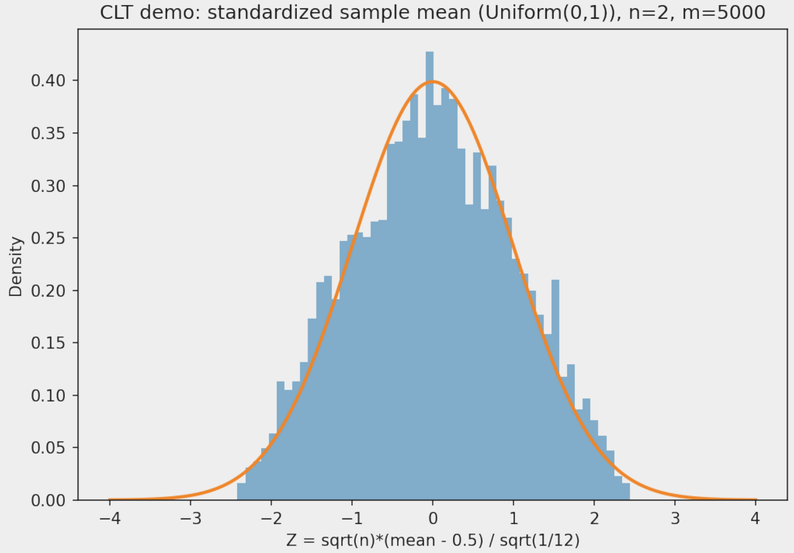
    # Plot: histogram (density) and standard normal overlay
    plt.figure(figsize=(7,5))
    plt.hist(Z, bins=50, density=True, alpha=0.6)
    plt.plot(xgrid, standard_normal_pdf(xgrid), linewidth=2)
    plt.title(f"CLT demo: standardized sample mean (Uniform(0,1)), n={n}, m={m}")
    plt.xlabel("Z = sqrt(n)*(mean - 0.5) / sqrt(1/12)")
    plt.ylabel("Density")
    plt.tight_layout()
    out_path = f"clt_uniform_hist_n{n}.png"
    plt.savefig(out_path, dpi=150)
    print(f"Saved: {out_path}")
    plt.show()
```

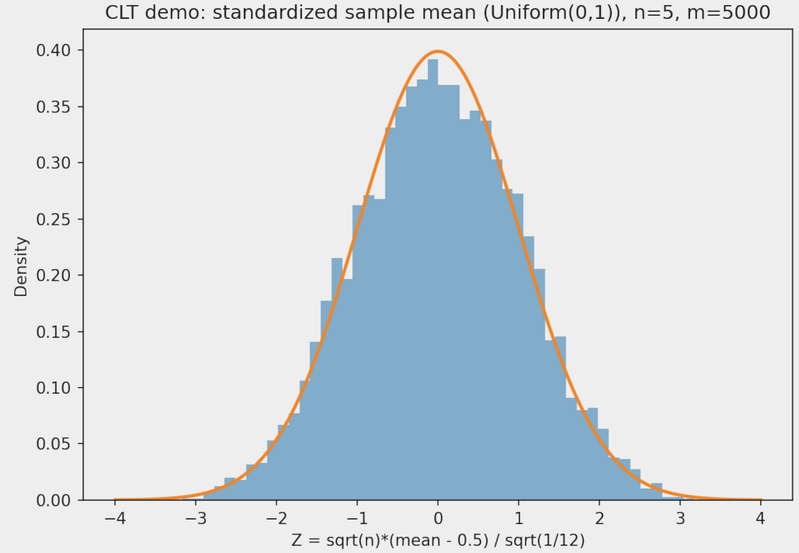
Inference - Week 3

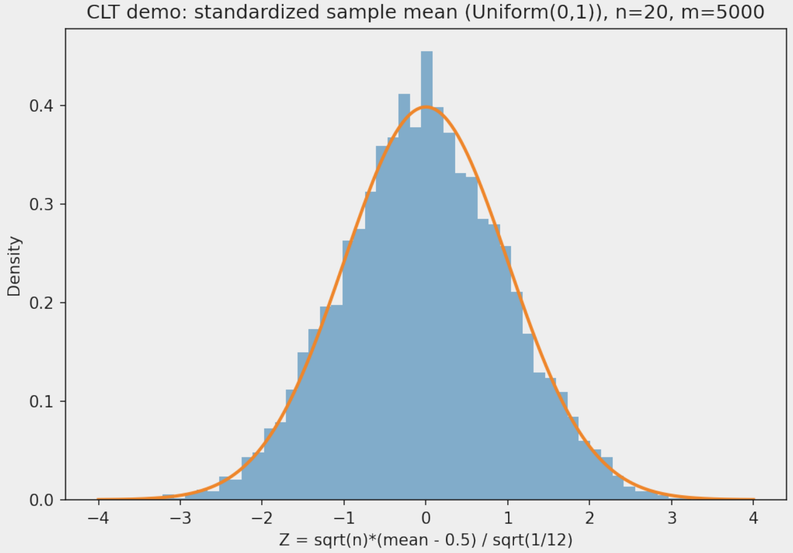


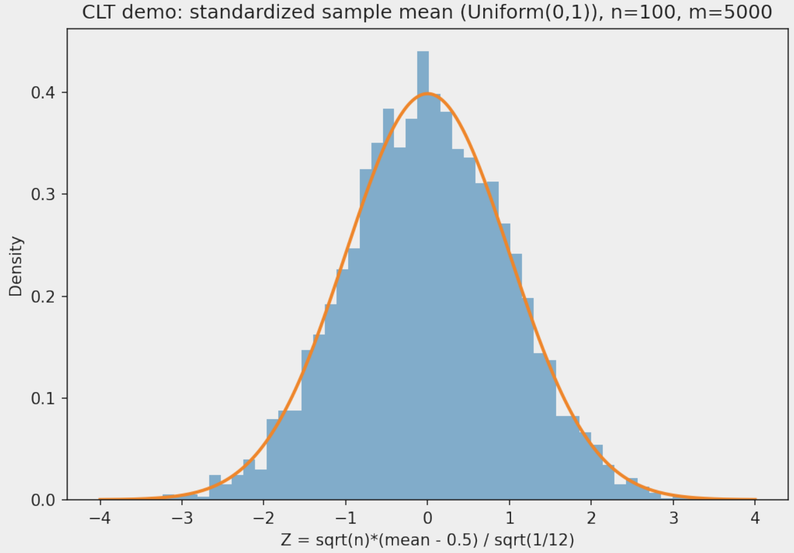
Inference - Week 3 CLT numerical example

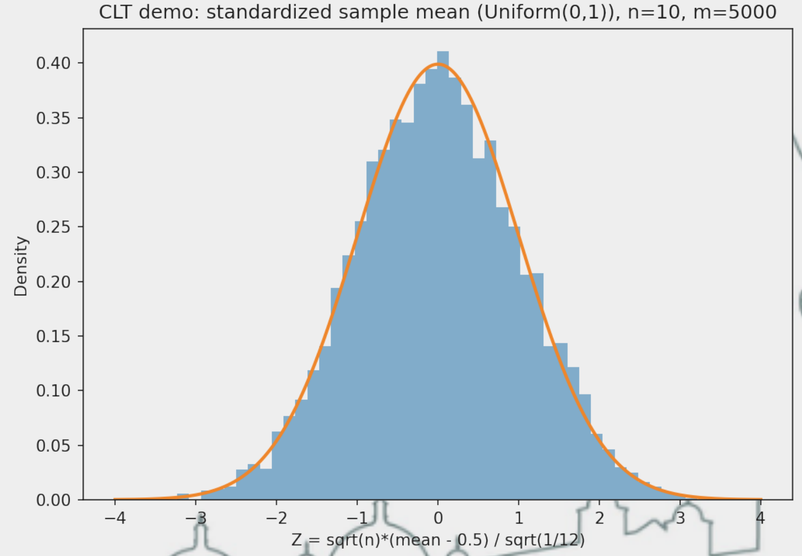












Inference - Week 3 CLT numerical example

```python
rows = []
xgrid = np.linspace(-0.1, 1.1, 400)

for n in n_list:
    # Simulate: shape (m, n), mean along axis=1
    samples = rng.random((m, n))
    sample_means = samples.mean(axis=1)
    Z = sample_means

    # Empirical stats
    z_mean = Z.mean()
    z_var = Z.var(ddof=1)

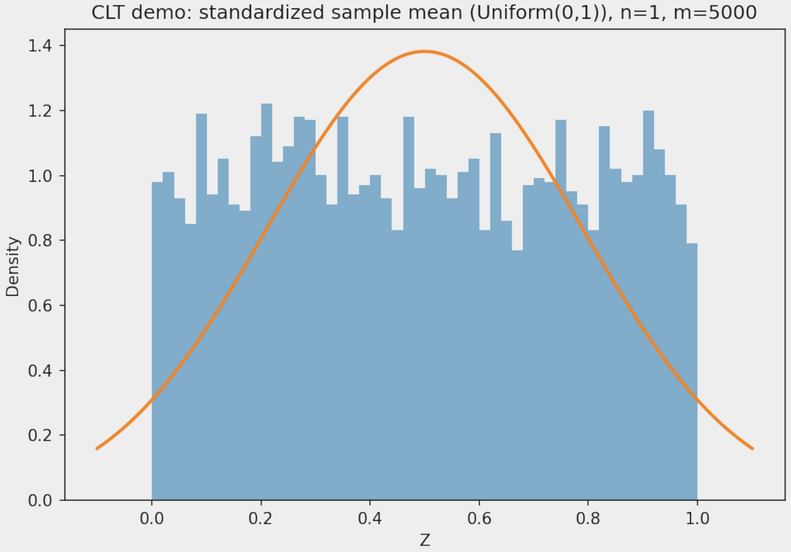
    rows.append({
        "n": n,
        "empirical_mean_Z": z_mean,
        "empirical_var_Z": z_var,
        "theoretical_mean_Z": 0.0,
        "theoretical_var_Z": 1.0
    })

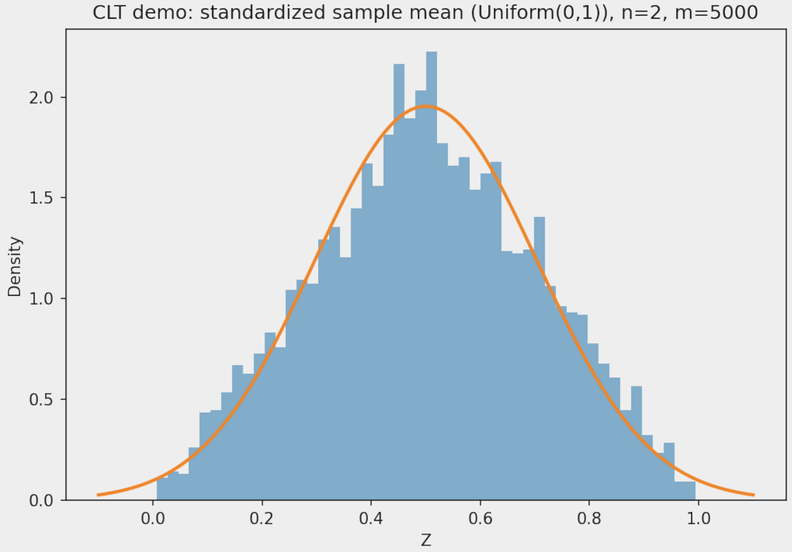
    # Plot: histogram (density) and standard normal overlay
    plt.figure(figsize=(7,5))
    plt.hist(Z, bins=50, density=True, alpha=0.6)
    plt.plot(xgrid, normal_pdf(xgrid, mu, sigma / n**0.5), linewidth=2)
    plt.title(f"CLT demo: standardized sample mean (Uniform(0,1)), n={n}, m={m}")
    plt.xlabel("Z")
    plt.ylabel("Density")
    plt.tight_layout()
    out_path = f"clt_uniform_unsc_hist_n{n}.png"
    plt.savefig(out_path, dpi=150)
    print(f"Saved: {out_path}")

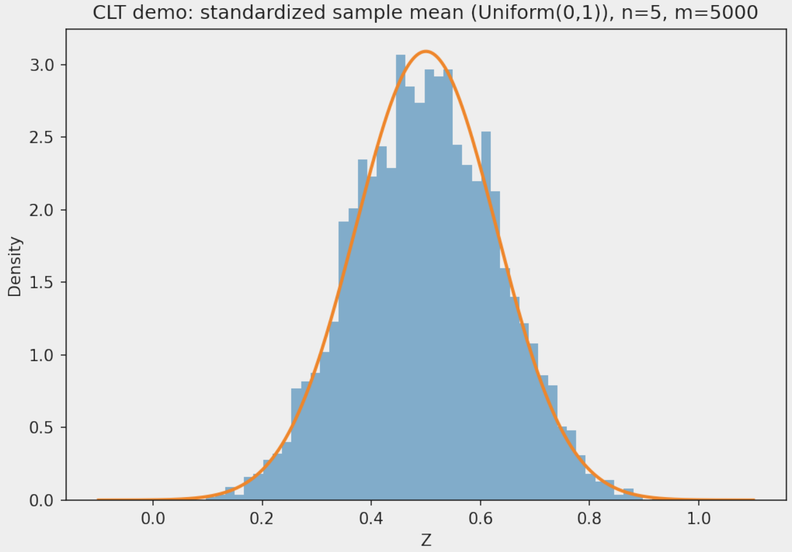
plt.show()
```

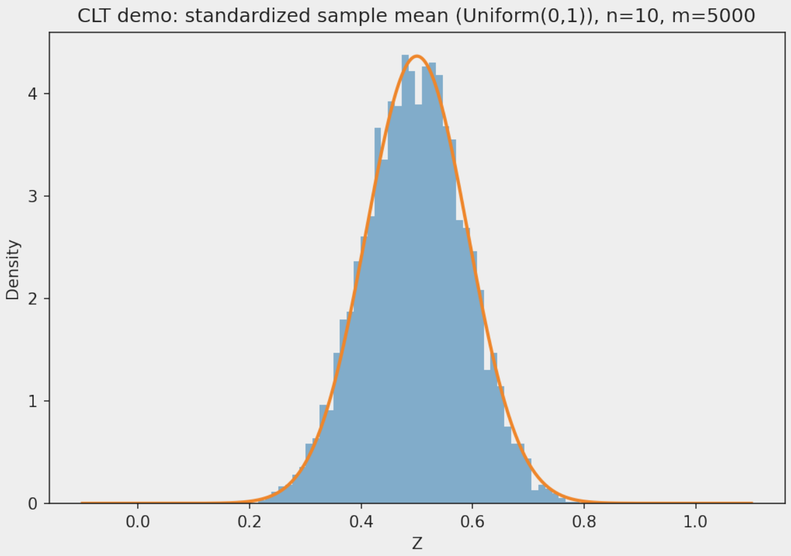


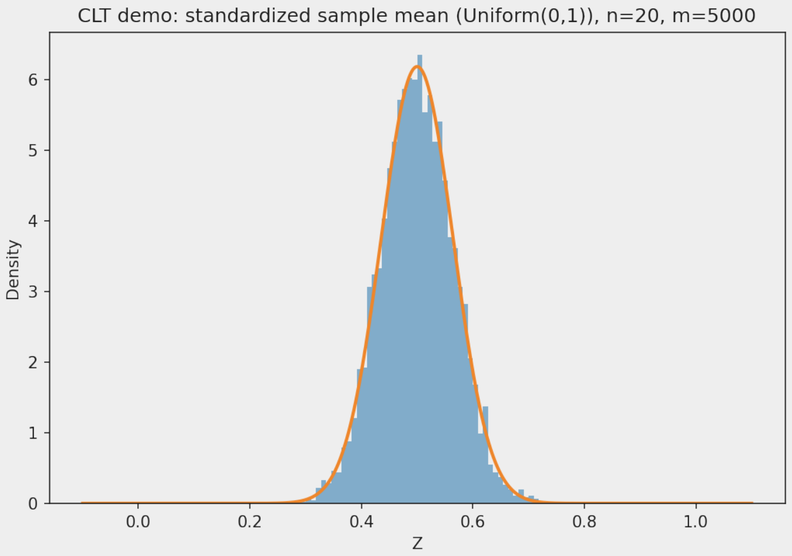
Inference - Week 3 CLT numerical example

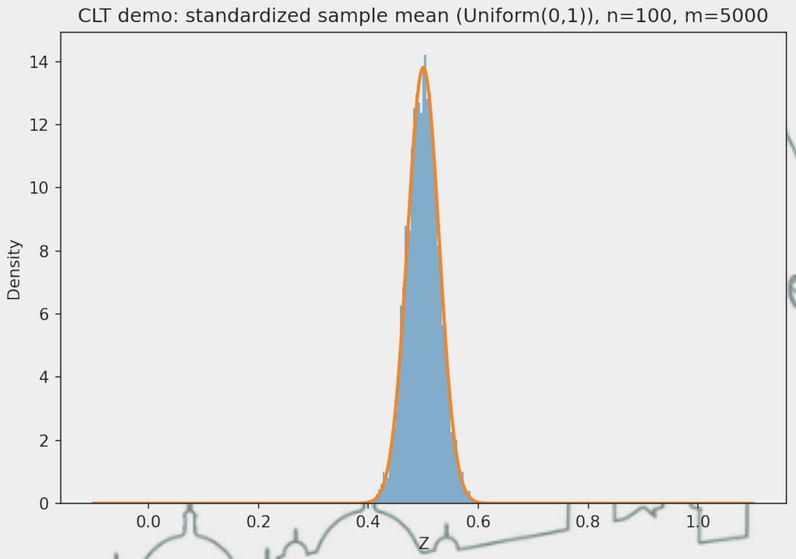












Inference - Week 3

Multivariate distributions

Multi-variate continuous distributions!

Let x and y be continuous random variables. We can have a 2D PDF:

$$P(x, y),$$

and we can work out the probability of some result:

$$\text{Prob}(a \leq x \leq b,\ c \leq y \leq d) = \int_a^b \int_c^d P(x, y)\, \mathrm{d}x\, \mathrm{d}y.$$

If x and y are independent then the PDF is seperable,

$$P(x, y) = P(x)\, P(y).$$



Inference - Week 3

1D marginal distribution

We marginalise (integrate) over the nusiance variable,

$$P(x) = \int \mathrm{d}y\, P(x, y).$$

1D conditional distribution

For example, what if we want to know the PDF of x for when $y = Y$, i.e., the conditional distribution,

$$P(x|y = Y) = P(x, y = Y),$$

noting that this will not be normalised. And so,

$$P(x|y = Y) = \frac{P(x, y = Y)}{\int \mathrm{d}x'\, P(x', y = Y)}.$$



Inference - Week 3

Other small points:

$$\langle f(x,y)\rangle = \int \mathrm{d}x \int \mathrm{d}y\, f(x,y)\, P(x,y),$$

$$\mathrm{Var}(x) = \left\langle (x - \langle x\rangle)^2 \right\rangle = \int \mathrm{d}x \int \mathrm{d}y\, (x - \langle x\rangle)^2\, P(x,y),$$

where,

$$\langle x\rangle = \int \mathrm{d}x \int \mathrm{d}y\, x\, P(x,y).$$

Covariance

Covariance can be defined:

$$\mathrm{Cov}(x,y) = \langle (x - \langle x\rangle)(y - \langle y\rangle)\rangle\,,$$

and Correlation:

$$\mathrm{Corr}(x,y) = \frac{\mathrm{Cov}(x,y)}{\sigma_x\, \sigma_y}$$



Inference - Week 3

In [8]:

```
1  import numpy as np
2  import seaborn as sns
3
4  penguins = sns.load_dataset("penguins")
5  penguins = penguins.sample(frac=1)
6  penguins.head()
7
```

Out[8]:

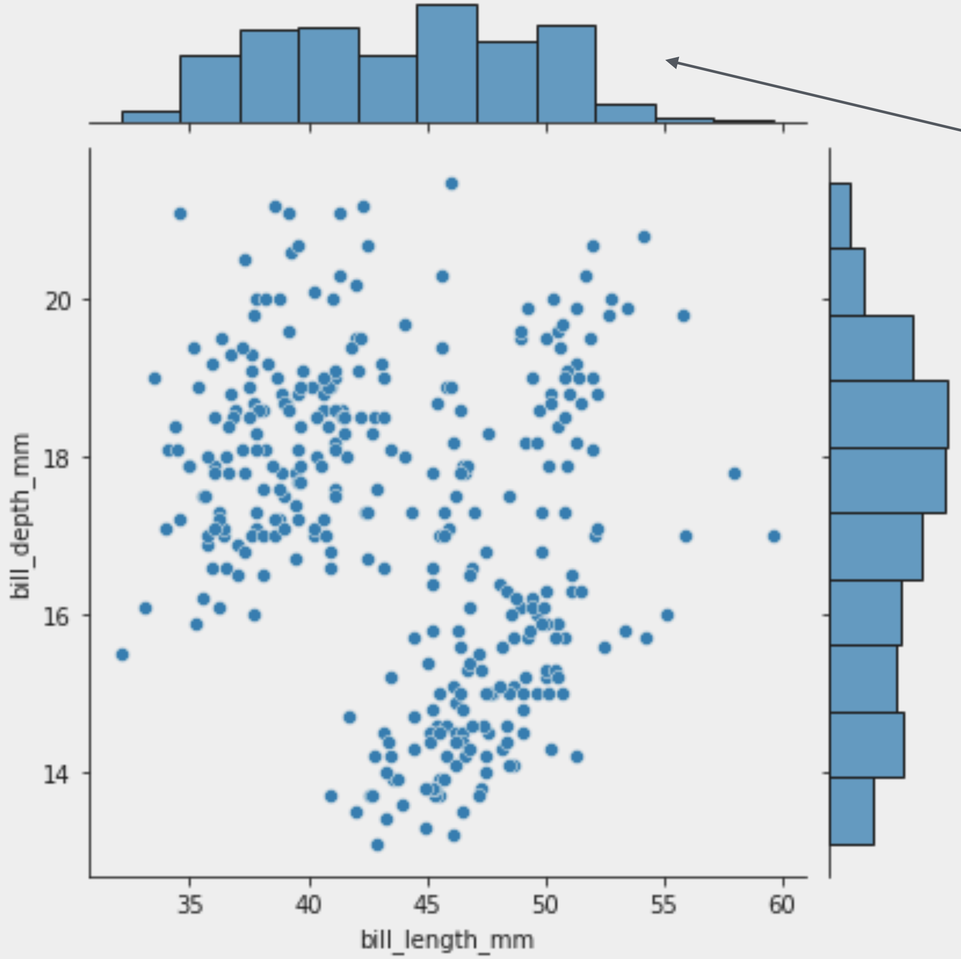
	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex
307	Gentoo	Biscoe	51.3	14.2	218.0	5300.0	Male
26	Adelie	Biscoe	40.6	18.6	183.0	3550.0	Male
341	Gentoo	Biscoe	50.4	15.7	222.0	5750.0	Male
69	Adelie	Torgersen	41.8	19.4	198.0	4450.0	Male
19	Adelie	Torgersen	46.0	21.5	194.0	4200.0	Male



Inference - Week 3

```
1 sns.jointplot(data=penguins, x="bill_length_mm", y="bill_depth_mm")
```

```
<seaborn.axisgrid.JointGrid at 0x7f80e85d07c0>
```

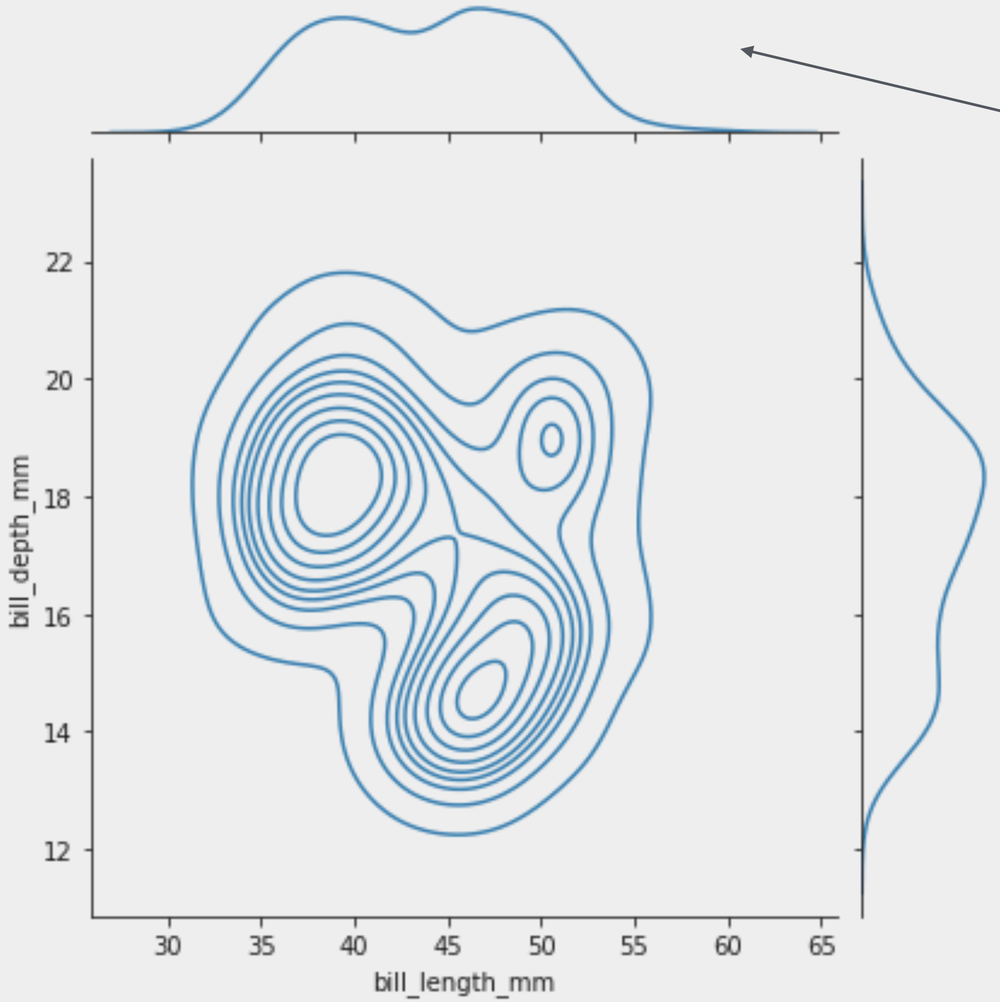


Marginals

Inference - Week 3

```
1 sns.jointplot(data=penguins, x="bill_length_mm", y="bill_depth_mm", kind='kde')
```

```
<seaborn.axisgrid.JointGrid at 0x7f80e3f524f0>
```



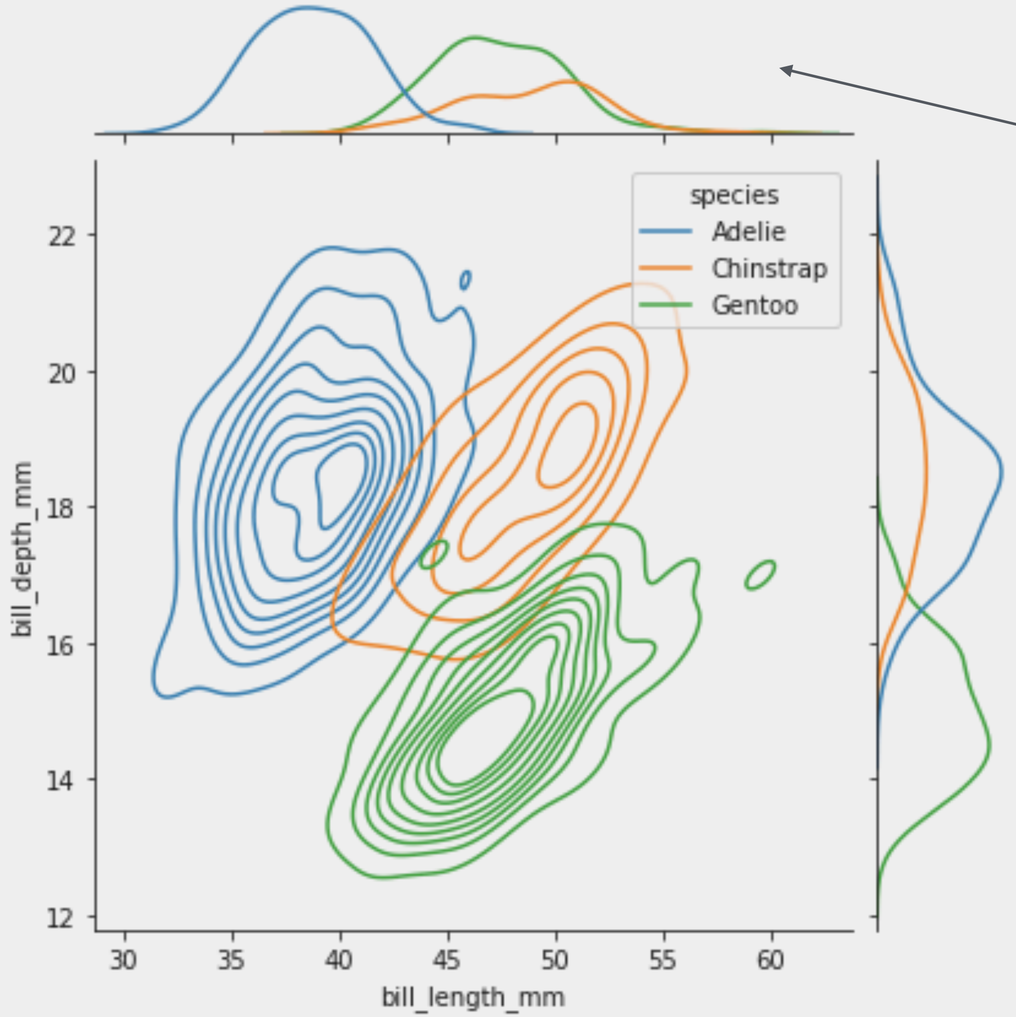
Marginals



Inference - Week 3

```
1 sns.jointplot(data=penguins, x="bill_length_mm", y="bill_depth_mm", hue="species", kind="kde")
```

```
<seaborn.axisgrid.JointGrid at 0x7f80e7850d30>
```



Conditional on species

Inference - Next time …

Bayesian Methods! Whoop

Quiet a lot of maths (sorry)

